

ACL Injury Mechanisms and Related Factors in Male and Female Carving Skiers: A Retrospective Study

Authors

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Key words

- ski injury
- carving ski
- ACL injury mechanism
- gender difference

Abstract



In recreational alpine skiing, ACL injury risk is 3 times greater in females. However, since the introduction of carving skis ACL injury risk seems to have decreased. No study has yet investigated the distribution of ACL injury mechanisms in male and female carving skiers. Therefore, the aim of the study was to investigate potential gender specific differences of ACL injury mechanisms and related factors among carving skiers. In total, 220 recreational carving skiers (59 males and 161 females) suffering from an ACL injury volunteered for this study. Demographic data, skiing ability, equipment related and environ-

mental factors, circumstances and causes for the fall, and type of fall (injury mechanisms) were collected by questionnaire. The forward twisting fall is the most reported ACL injury mechanism in both gender ($p=0.672$) accounting for 54% of all injuries, although male and female skiers differed significantly with regard to circumstances of fall ($p=0.001$) and actions when ACL injury occurred ($p=0.04$). Bindings not releasing at the time point of accident occurred 2.6 times more with females than with males ($p=0.005$). The forward twisting fall seems to have become the dominant ACL injury mechanism both in male and female recreational skiers since the introduction of carving skis.

Introduction



In alpine skiing, the knee joint accounts for about one-third of all injuries in recreational and competitive skiers [2,8,28]. The anterior cruciate ligament (ACL) is affected in approximately 50% of serious knee injuries [5,27,33] and in about 20% of all skiing injuries [16]. However, there are distinctive gender differences in rates of knee injuries. Female recreational and competitive skiers have twice the knee injury incidence of male skiers and the ACL injury risk is 3 times greater in female skiers [2,4,6,33].

With increasing popularity of the short and shaped carving ski, an increase in the knee injury rate has been expected because of an increased potential for catching an outside edge [15]. However, Bartscher et al. [2] showed that the gender specific knee injury rate remained constant with the introduction of carving skis while the overall injury rate has decreased by 9%. In addition, Ettlinger et al. [7] reported a significant decrease in the risk of sustaining an ACL injury in recent years as carving skis became more popular. While ACL injury mechanisms on traditional skis are

well studied [1,14,35], only few data are available on ACL injury mechanisms in carving skiers. A recent study has shown the forward twisting fall to be the most common ACL injury mechanism in female carving skiers [29]. However, no study has yet investigated gender differences concerning the distribution of ACL injury mechanisms. Therefore, the aim of this study was to investigate gender specific differences of ACL injury mechanisms and related factors among carving skiers.

Materials and Methods



Subjects

This study was conducted as a retrospective questionnaire based study during the 2 winter seasons 2008/2009 and 2009/2010 in 2 Austrian ski injury clinics. Inclusion criteria were non-contact ACL injury and the use of carving skis in recreational skiers of both sexes. With regard to skiing ability, expert skiers (ski racers and ski instructors) were excluded. MRI was used for the diagnosis of partial or total tear of the ACL. In

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total, 220 recreational carving skiers (59 males and 161 females) suffering from an ACL injury volunteered for this study representing 20% of all ACL injuries diagnosed at the enrolled ski clinics during the 2 winter seasons. The random recruitment of patients was dependent upon organizational aspects at the clinics (availability of rooms and personnel) and willingness of patients to volunteer. More than 90% of invited subjects agreed to participate. The study was performed in conformity with the ethical standards of the 1975 Declaration of Helsinki and with the ethical standards of this journal [10]. Informed consent was obtained from all subjects prior to the beginning of this research.

Questionnaire

Demographic data, skiing ability, equipment related and environmental factors, circumstances and causes for the fall, and type of fall (injury mechanisms) were collected by questionnaire within 2 days after injury with a physician present.

We recorded the time of skiing before the occurrence of injury (≤ 2 vs. > 2 h). Skiing ability was classified into 4 categories termed expert (racer, ski instructor), advanced (can ski any run in any condition), intermediate (ski on medium runs under good conditions; have difficulties controlling skis on poorly groomed runs or in bad weather) or beginner (ski only easy runs which are well groomed) according to the study by Sulheim et al. [34]. Additionally, according to the study of Burtcher et al. [3] number of falls (0 vs. 1–3 vs. > 3 falls) on the day before the occurrence of the ACL injury was recorded.

Equipment related factors consisted of ski length of the carving skis, ski length to height ratio, gear origin (own, ski shop rental, borrowed gear from family members/friends), date of last binding adjustment (≤ 1 year vs. > 1 year), and whether the ski binding has released at the time of accident or not. If an injured person using hired skis did not/could not answer the question about the date of the last binding adjustment, we assumed that the last binding adjustment had been made when hiring skis in a professional Austrian ski shop, i.e., ≤ 1 year ago.

Environmental factors included snow conditions (fresh snow, grippy, icy, slushy/soft), difficulty of the downhill slope (easy, moderate, hard), and weather (sunny, overcast, snowfall).

Circumstances of the fall were subdivided into: (a) after jumping, (b) caught an edge, (c) ski slide out or away/ lost balance, (d) accidental binding release, and (e) others. Additionally, we asked subjects whether the ACL injury occurred while (a) executing a turn, (b) going straight, (c) landing after a jump, or (d) don't know.

According to the study by Järvinen et al. [14], patients recalled their type of fall by observing pictures of the most typical ACL injury mechanisms in skiing. 4 categories for the direction of the fall were classified: (a) forward fall with body rotation, (b) forward fall without body rotation, (c) backward fall with body rotation, and (d) backward fall without body rotation.

Statistics

Data are presented as means and absolute or relative frequencies. Unpaired t-tests and Mann-Whitney-U-tests, as appropriate, were used to compare cases and controls with regard to age, height, weight, and BMI as well as absolute and relative ski length. Differences in frequencies were evaluated by Chi-square-tests. Odds ratios (OR) and 95% confidence intervals (CI) were calculated to investigate the gender-specific lack of binding release at the time point of accident. All P-values were two-tailed and values below 0.05 were considered to indicate statistical significance.

Results

Demographic data are shown in **Table 1**. Male and female recreational skiers suffering from ACL injury did not differ in age or average time of skiing before the event of the injury occurred. However, males were significantly taller, heavier and had a higher BMI ($p < 0.001$). In addition, males were more likely to be Austrian nationals and more likely advanced skiers compared to

	Males N = 59	Females N = 161	p-value
demographic data			
age, years	43.6 [11.5], 17–75	42.25 [10.31], 14–72	0.395
height, cm	181.4 [6.7], 170–188	166.7 [5.9], 151–182	< 0.001
weight, kg	85.6 [14.6], 55–129	66.3 [10.0], 47–103	< 0.001
BMI	26.0 [3.7], 18.6–36.3	23.9 [3.5], 16.9–40.2	< 0.001
nationality			
Austrian	12 (20.3)	12 (7.5)	
German	39 (66.1)	117 (72.7)	
others	8 (13.6)	32 (19.9)	0.020
time of skiing before injury			
≤ 2 h	27 (46.6)	92 (57.1)	
> 2 h	31 (53.4)	69 (42.9)	0.165
missing data	1		
skiing ability			
skill level			
advanced	40 (67.8)	76 (47.2)	
intermediate	18 (30.5)	71 (44.1)	
beginner	1 (1.7)	14 (8.7)	0.014
falls before injury			
0 falls	50 (86.2)	124 (80.0)	
1–3 falls	8 (13.8)	28 (18.1)	
> 3 falls	0 (0)	3 (1.9)	0.411
missing data	1	6	

Data are presented as means [$\pm$ SD], range or frequencies (percentages)

Table 1 Demographic data and skiing ability in male and female ACL injured carving skiers.

females ($p < 0.05$). No difference was shown within the number of falls before the ACL injury occurred.

Equipment related and environmental factors are presented in **Table 2**. While males used on average 15 cm longer carving skis ($p < 0.001$), ski length to height ratio (93%) was identical within both genders ($p > 0.05$). No significant difference was shown with regard to gear origin. A trend was found ($p = 0.082$)

for females using more frequently a newly adjusted binding (≤ 1 year) than males. In total, 63.8% of males and 82% of females reported that their binding did not release at the time point of accident ($p = 0.005$). An OR of 2.6 (95% CI: 1.32–5.05) was calculated. No gender differences were shown with regard to snow conditions, slope difficulty, and weather ($p > 0.05$).

	Males N = 59	Females N = 161	p-value
equipment related factors			
ski length, cm	169.4 [10.6], 130–191	154.7 [8.1], 130–173	<0.001
missing data	4	34	
ski length to height ratio, %	93.3 [4.6], 70.3–100.6	92.8 [4.3], 80.3–103	0.491
missing	4	34	
gear origin			
own	43 (72.9)	105 (65.2)	
ski shop rental	15 (25.4)	52 (32.3)	
borrowed gear	1 (1.7)	4 (2.5)	0.558
binding adjustment			
≤ 1 year (%)	39 (66.1)	125 (77.6)	
> 1 year (%)	20 (33.9)	36 (22.4)	0.082
binding released at accident			
yes	21 (36.2)	29 (18.0)	
no	37 (63.8)	132 (82.0)	0.005
missing	1		
environmental factors			
snow conditions			
fresh snow	15 (25.4)	30 (19.2)	
grippy	25 (42.4)	63 (40.4)	
icy	13 (22.0)	44 (28.2)	
slushy/soft	6 (10.2)	19 (12.2)	0.662
missing data		5	
slope difficulty			
easy (blue)	14 (23.7)	15 (27.0)	
moderate (red)	37 (62.7)	42 (62.3)	
hard (black)	8 (13.6)	11 (10.7)	0.782
missing data		2	
weather			
sunny	33 (55.9)	101 (66.0)	
overcast	18 (30.5)	42 (27.5)	
snow fall	8 (13.6)	10 (6.5)	0.192
missing data		8	

Data are presented as means [$\pm$ SD], range or frequencies (percentages)

Table 2 Equipment related and environmental factors in male and female ACL injured carving skiers.

	Males N = 59	Females N = 161	p-value
circumstances of fall			
after jumping, N (%)	4 (7.3)	3 (2.0)	0.001
caught an edge, N (%)	25 (45.5)	37 (25.2)	
ski slid out or away/lost balance, N (%)	19 (34.5)	86 (58.5)	
accidental binding release, N (%)	2 (3.6)	0 (0)	
others, N (%)	5 (9.1)	21 (14.3)	
missing data	4	14	
action when ACL injury occurred			
executing a turn, N (%)	41 (75.5)	94 (63.9)	0.040
going straight, N (%)	5 (9.1)	23 (15.6)	
landing after a jump, N (%)	3 (5.5)	1 (0.7)	
don't know, N (%)	6 (10.9)	29 (19.7)	
missing data	4	14	
type of fall			
forward fall with rotation	30 (54.5)	78 (54.2)	0.672
forward fall without rotation	6 (10.9)	15 (10.4)	
backward fall with rotation	14 (25.5)	44 (30.6)	
backward fall without rotation	5 (9.1)	7 (4.9)	
missing data	4	17	

Data are presented as frequencies (percentages)

Table 3 Circumstances and types of falls in male and female ACL injured carving skiers.

Circumstances and types of falls are shown in **Table 3**. Gender differences were found regarding circumstances of fall ($p=0.001$) and action when ACL injury occurred ($p=0.04$). Males reported more often catching an edge as cause for the accident while females reported more often sliding out and losing balance. Males injured their ACL more often during turning than females while female skiers injured their ACL more often when going straight compared to male skiers.

No significant gender differences regarding the distribution of injury mechanism were detected ($p=0.672$). The forward twisting fall was the most reported injury mechanism associated with ACL injury in male and female carving skiers, followed by the backward fall with body rotation. In total, 32.4% of male and female skiers with a forward twisting fall reported that the binding released at the time point of accident compared to 7.8% of skiers with other falls ($p<0.001$). An OR of 5.7 (95% CI: 2.38–13.75) was calculated.

Discussion

The main finding of this study was that there are no gender differences of the distribution of ACL injury mechanisms among carving skiers. The forward twisting fall is the most reported ACL injury mechanism in both male and female carving skiers. However, males and females differed significantly with regard to circumstances of fall and actions when ACL injury occurred. Bindings not releasing at the time point of accident occurred 2.6-fold more with females compared to males although more female skiers reported an actual binding adjustment by a professional ski service (≤ 1 year) compared to male skiers ($p=0.082$). No significant gender differences were detected with regard to environmental factors.

Regarding circumstances leading to the accident, males injured their ACL more often while turning and more often by catching an edge compared to females. In contrast, females fell more often when going straight and because their ski slid out. These differences might be due to the fact that males are more skilled, skied on average faster and showed a higher risk taking behaviour than females [30]. In accordance, our results showed significantly more advanced male skiers than females. However, in our previous study [29] catching an edge was the most reported cause of accident in ACL injured female carving skiers. The contrasting results of the present study might be due to the higher number of ACL injured females, to different snow and weather conditions and to the fact that with regard to skiing ability, expert skiers (ski racers and ski instructors) were excluded.

In alpine skiing 3 main non-contact ACL injury mechanisms have been identified: forward twisting fall ("valgus-external rotation"), boot-induced anterior drawer, and the backward twisting fall ("flexion-internal rotation" or "phantom foot") [1,14,25,27,28]. While some researchers stated the backward twisting fall to be the most common ACL injury mechanism [1,17,25] others believe that the forward twisting fall could be the dominant ACL injury mechanism in recreational skiers [9,14,35]. However, these studies are widely based on ACL-injured skiers using so called traditional or conventional (long and unshaped) skis. Regarding female carving skiers, the current results are well in accordance with our prior study [29] from the winter season 2007/2008 showing the forward twisting fall to be the most reported injury mechanism with 51% followed by the backward twisting fall with 29%. In the present study, we found no gender

difference in the distribution of ACL injury mechanisms although skill level differed significantly. In comparison, Greenwald and Toelcke [9] found that the forward twisting fall was the most common ACL injury mechanism ($>50\%$) in experienced non-competitive skiers while in beginner and competitive skiers a higher number of knee injuries were associated with backward twisting falls.

There are some indications that the incidence of ACL injuries in alpine skiing is affected by the introduction of the short and shaped carving skis. First of all, Ettlinger et al. [7] reported a significant decrease in the risk of sustaining an ACL injury in previous years when carving skis became more popular. In addition, other studies revealed even a lower risk for sustaining an ACL injury when skiing with carving skis compared to traditional skis [20,23]. In a study conducted in a German ski clinic, carving skiers suffered significantly less knee injuries (37.6% vs. 41.8%) and less ACL ruptures (11.4% vs. 14.3%) compared to those with traditional skis [20]. We assume that this result may be partly related to changes in the distribution of ACL injury mechanisms with the introduction of the short and shaped carving ski [29]. Carving skis are recommended to be 10–15 cm shorter than traditional skis [11]. While the length of a carving ski is usually adapted to the height of the skier's chin or nose, traditional ski's length was usually about 10 cm above the skier's body height. Relative ski length in this study was 93% of body height which is well in accordance with our previous finding [29]. Johnson et al. [17] reported that ski length less than 75% of the skier's height was unlikely to be associated with an ACL injury. In addition, in another study by Johnson et al. [18] so called ski boards with a usual length of 100 cm (which would correspond to 59–65% body height in our study population) or less have a much lower ACL injury risk compared to traditional skis. These authors believe that the majority of ACL injuries are caused by a backward twisting fall where the long tail of a traditional ski act as a lever applying a unique combination of twisting and bending loads to the flexed or hyper-flexed knee, the so called "phantom foot" [18]. They stated that the short design of a ski board effectively amputates most of the "phantom foot", thereby practically eliminating the risk of ACL injury in ski boarders [18]. Therefore, one can assume the shorter the ski the lower the ACL injury risk. In addition, Burtscher et al. [2] found that the introduction of carving skis was associated with increased proportions of shoulder/back and head injuries probably due to the increased incidence of backward twisting falls. However, an increase of backward twisting falls caused by short carving skis [2] and a decrease of ACL injuries [7] at the same time might only be possible when the incidence of backward twisting falls causing ACL injuries ("phantom foot"-mechanism) has decreased. Therefore, it seems likely that the shorter tail of a carving ski compared to traditional skis caused a change in the distribution of ACL injury mechanisms from the backward twisting fall to the forward twisting fall. In addition, although ski bindings are not primarily designed to protect the knee joint, release function of the current bindings is at least provided for the forward twisting fall [12]. Thus, during a forward twisting fall there seems to be a higher chance of a binding release without an injury. In the present study, 32.4% of male and female skiers with a forward twisting fall reported that the binding released at the time point of accident compared to 7.8% of skiers with other falls ($p<0.001$). Importantly, we observed a strong gender difference with regard to binding release. The lack of binding release in 82% of females and 64% of males in this study is well in accordance with the

results by Greenwald and Toelcke [9] who reported a lack of binding release in 88% of women's ACL ruptures compared to 67% for men. Also, in our prior study, about 84% of ACL-injured female carving skiers reported a lack of binding release at the time point of accident [29]. To date, release values of the bindings according to the ISO 11088 standard [13] were selected mainly using the weight of the skiers without considering other gender specific differences. Werner and Willis [36] showed that muscle strength is highly correlated with the ability to release the binding in a self-release test which is a comparable movement to the forward twisting fall mechanism. The weight-to-strength ratio, however, is negatively influenced by the higher fat mass in females [32]. The importance of binding setting in female skiers has recently been discussed by LaPorte et al. [22]. In a case-control study, they found that lower binding release values in female skiers set 15% lower than those recommended by the ISO 11088 standard [13] would clearly reduce knee injuries in these persons. No increase in injuries from inadvertent binding release through reduced binding settings has been found [22]. However, these findings mean an association, and not definitely "cause and effect" relationship, because another study showed a decrease in ACL injury risk without reducing binding settings [7]. It seems possible that the use of carving skis altered the distribution of injury mechanisms contributing to a decrease in the risk of ACL injury in male and female carving skiers.

Limitations

Although we do not have any indication, a possible selection bias can not be excluded entirely. The observed gender ratio supports the assumption that the investigated sample is representative for the whole ACL injured population of carving skiers. Unfortunately, compared to the backward twisting fall and the boot induced anterior drawer mechanism, no video evidence for the forward twisting fall in recreational skiing has been found [1]. However, studies from other sports also used retrospective interviews to describe ACL injury mechanisms [19,21]. In addition, Olsen et al. [26] and Shimokochi and Shultz [31] reported that observational analyses of ACL injury mechanisms using videos are consistent with retrospective interviews of injured people. Additionally, one study found that video analysis alone was not able to detect the instant of ACL rupture in half of injured ski racers [24]. In general, the potential for injured skiers to forget exactly what happened and give erroneous answers has to be mentioned.

Conclusion

Based on these findings with the introduction of carving skis the forward twisting fall seems to have become the dominant ACL injury mechanism with no differences in male and female recreational skiers.

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